



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE602 / 20EE602 - SWITCHGEAR AND PROTECTION

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define a fuse and explain the issues concerning material selection for fuse elements.	5	L*2	1	1
	b) With the neat sketch describe the working principle of a liquid fuse.	10	L1	1	1
	c) List the advantages of Neutral grounding.	5	L2	1	1
2.	a) Explain the current chopping phenomenon in a circuit breaker with a neat wave form.	6	L2	2	1,2
	b) Explain the concept of resistance switching.	6	L2	2	1,2
	c) Explain the time-current characteristics of a fuse.	8	L1	1	1
3.	a) With the neat sketch explain the construction and working principle of HRC fuse. Mention its advantages.	10	L1	1	1
	b) Explain clearly with figure the current interruption in a.c. circuit breaking. Hence define the terms restriking voltage and recovery voltage.	10	L2	2	1,2
Unit – II					
4.	a) With the neat diagram explain the operation of vacuum circuit breakers. State the advantages of vacuum circuit breakers.	10	L2	3	1
	b) Discuss the operating principle of SF6 circuit breaker. What are its advantages?	10	L2	3	1
5.	a) Explain the working of air blast circuit breaker with reference to i) Axial blast ii) Cross blast	10	L2	3	1
	b) Explain the procedure adopted in unit test and synthetic testing of circuit breaker.	10	L2	3	1
6.	a) Explain in brief basic requirements of protective relaying.	12	L1	4	1
	b) Explain with the help of neat block diagram the construction and working of non directional induction type over current relay.	8	L2	4	1,2
Unit – III					
7.	a) Describe in brief different protective schemes provided for protection of generators.	12	L2	5	1,2
	b) Draw and explain the based differential protection scheme of Star-Delta transformer.	8	L2	5	1,2
8.	a) Describe in brief the protection of motors against unbalance and single phasing.	10	L2	5	1,2
	b) List and explain the various abnormal operating conditions against which a large induction motor has to be protected	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
